

Probability and Random Variables – Midterm Examination

Time: 120 minutes Total: 100 points

Instructions: Please answer all questions in the spaces provided. Show your work for calculation problems.

Part I: Multiple Choice (10 points)

Choose the best answer for each question. (2 points each)

1. Which of the following statements about events and probability is correct?
 - A. If A and B are mutually exclusive and $\mathbb{P}(A) > 0, \mathbb{P}(B) > 0$, then they are independent.
 - B. If A and B are independent, then they must be mutually exclusive.
 - C. If $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$, then A and B are independent.
 - D. If A and B are mutually exclusive, then $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$ holds only when $\mathbb{P}(A \cap B) = 0$.
2. Let X be a continuous random variable with cumulative distribution function (CDF) F_X . Which of the following is always true?
 - A. F_X is left-continuous.
 - B. $F_X(x) = \mathbb{P}(X < x)$ for all $x \in \mathbb{R}$.
 - C. F_X is monotonically decreasing.
 - D. $\lim_{x \rightarrow -\infty} F_X(x) = 0$.
3. The probability mass function (PMF) of a Poisson random variable X with parameter $\theta > 0$ is:
 - A. $f_X(k) = \frac{e^{-\theta}\theta^k}{k!}, \quad k = 0, 1, 2, \dots$
 - B. $f_X(k) = \binom{n}{k}\theta^k(1-\theta)^{n-k}, \quad k = 0, \dots, n$
 - C. $f_X(k) = \theta(1-\theta)^{k-1}, \quad k = 1, 2, \dots$
 - D. $f_X(k) = \frac{1}{\sqrt{2\pi}\sigma}e^{-(k-\mu)^2/(2\sigma^2)}$
4. If $X \sim N(\mu, \sigma^2)$, then which transformation yields a standard normal random variable?
 - A. $\frac{X - \mu}{\sigma}$
 - B. $\frac{X + \mu}{\sigma}$
 - C. $\frac{X - \sigma}{\mu}$
 - D. $\frac{X + \sigma}{\mu}$
5. For a random vector (X, Y) with joint probability density function $f(x, y)$, the marginal PDF of X is given by:
 - A. $\int_{-\infty}^{\infty} f(x, y) dy$

- B. $\int_{-\infty}^{\infty} f(x, y) dx$
- C. $\frac{f(x, y)}{\int_{-\infty}^{\infty} f(x, y) dy}$
- D. $\frac{f(x, y)}{\int_{-\infty}^{\infty} f(x, y) dx}$

Part II: Fill in the Blanks (10 points)

Complete each statement with the correct expression. (2 points each blank)

1. Suppose A and B are events with $\mathbb{P}(A) = 0.4$, $\mathbb{P}(B) = 0.5$, $\mathbb{P}(A \cap B) = 0.2$. Then $\mathbb{P}(A \cup B) = \underline{\hspace{1cm}}$ and $\mathbb{P}(A | B) = \underline{\hspace{1cm}}$.
2. Let X have probability density function $f(x) = c(1 - x^2)$ for $-1 < x < 1$ and 0 otherwise. The constant c equals $\underline{\hspace{1cm}}$.
3. If $X_1, \dots, X_n$ are i.i.d. $N(\mu, \sigma^2)$, then the sample mean $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ follows the distribution $\underline{\hspace{1cm}}$.
4. Bayes' theorem states: $\mathbb{P}(A_i | B) = \frac{\mathbb{P}(B | A_i)\mathbb{P}(A_i)}{\underline{\hspace{1cm}}}$.
5. A random variable X has a Gamma distribution with shape $a > 0$ and scale $b > 0$. Its PDF is $f_X(x) = \underline{\hspace{1cm}}$ for $x > 0$, where $\Gamma(\cdot)$ is the Gamma function.

Part III: Short Answer (20 points)

Provide concise but complete answers. (5 points each)

1. Define a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Explain what a σ -algebra $\mathcal{F}$ is and state its closure properties.
2. State the three defining properties of a cumulative distribution function (CDF) $F : \mathbb{R} \rightarrow [0, 1]$. Then, for a continuous random variable, how can you obtain the PDF from the CDF?
3. Let X and Y be continuous random variables with joint PDF $f_{X,Y}(x, y)$. State the condition for X and Y to be independent in terms of $f_{X,Y}$ and the marginal PDFs f_X , f_Y . Additionally, write the formula for the conditional PDF $f_{X|Y}(x | y)$ when $f_Y(y) > 0$.
4. Write the probability mass function of a multinomial distribution $\text{Multinomial}(n, \boldsymbol{\theta})$ with k categories. What is the marginal distribution of a single component X_i ?

Part IV: Calculation Problems (40 points)

Show all steps.

1. (10 points) **Conditional probability and Bayes.** A medical test for a disease has the following probabilities:
 - Disease prevalence: $\mathbb{P}(D) = 0.01$ (the probability a randomly selected person has the disease).
 - Sensitivity: $\mathbb{P}(+ | D) = 0.95$ (test positive given disease).
 - False positive rate: $\mathbb{P}(+ | D^c) = 0.05$ (test positive given no disease).

If a person tests positive, what is the probability that they actually have the disease? Compute $\mathbb{P}(D \mid +)$.

2. (10 points) **Joint and marginal distributions.** Let (X, Y) have joint PDF

$$f(x, y) = \begin{cases} x + y, & 0 \leq x \leq 1, 0 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

- Find the marginal PDFs $f_X(x)$ and $f_Y(y)$.
 - Compute $\mathbb{P}(X > 0.5, Y \leq 0.5)$.
 - Are X and Y independent? Justify your answer.
 - Determine the conditional PDF $f_{X|Y}(x \mid y)$ for $0 < y < 1$.
3. (10 points) **Transformation of a random variable.** Let X be an exponential random variable with rate $\lambda > 0$, i.e., $f_X(x) = \lambda e^{-\lambda x}$ for $x > 0$. Find the probability density function of $Y = \sqrt{X}$ using the transformation method (strictly monotonic case). Specify the support of Y .
4. (10 points) **Bivariate transformation.** Suppose X_1 and X_2 are independent exponential random variables with common rate $\lambda > 0$, so their joint PDF is $f_{X_1, X_2}(x_1, x_2) = \lambda^2 e^{-\lambda(x_1 + x_2)}$ for $x_1 > 0, x_2 > 0$. Define the transformation

$$Y_1 = X_1 + X_2, \quad Y_2 = \frac{X_1}{X_1 + X_2}.$$

- Show that the inverse transformation is $X_1 = Y_1 Y_2$, $X_2 = Y_1(1 - Y_2)$ and find the Jacobian determinant $\left| \frac{\partial(x_1, x_2)}{\partial(y_1, y_2)} \right|$.
- Derive the joint PDF $f_{Y_1, Y_2}(y_1, y_2)$ and specify its support.
- Are Y_1 and Y_2 independent? Explain.

Part V: Case Studies (20 points)

1. (10 points) **Spam filter application.** In an email system, three categories are defined: A_1 = spam, A_2 = low priority, A_3 = high priority. Prior probabilities are $\mathbb{P}(A_1) = 0.7$, $\mathbb{P}(A_2) = 0.2$, $\mathbb{P}(A_3) = 0.1$. Let B be the event that an email contains the word “free”. Conditional probabilities are $\mathbb{P}(B \mid A_1) = 0.9$, $\mathbb{P}(B \mid A_2) = 0.01$, $\mathbb{P}(B \mid A_3) = 0.01$.
- Compute the probability that an email contains the word “free” (i.e., $\mathbb{P}(B)$) using the law of total probability.
 - Given that an email contains the word “free”, find the posterior probability that it is spam, i.e., $\mathbb{P}(A_1 \mid B)$.
 - Also compute $\mathbb{P}(A_3 \mid B)$. Based on these probabilities, which category is most likely for an email containing “free”?
2. (10 points) **Inverse transform method.** The probability integral transform (Theorem 1.15 in the notes) states: if $U \sim \text{Uniform}(0, 1)$ and F is any CDF, then $X = F^{-1}(U)$ has CDF F , where $F^{-1}(u) = \inf\{x : F(x) \geq u\}$.
- Consider the exponential distribution with rate $\lambda > 0$, whose CDF is $F(x) = 1 - e^{-\lambda x}$ for $x \geq 0$. Find its quantile function $F^{-1}(u)$ for $u \in (0, 1)$.
 - Explain how you would generate 1000 i.i.d. exponential($\lambda = 2$) random variables using a computer that only produces i.i.d. Uniform(0, 1) numbers.

- (c) Derive the PDF of $X = F^{-1}(U)$ explicitly when $U \sim \text{Uniform}(0,1)$ and F is the exponential CDF given above, confirming that it matches the exponential PDF.

End of Examination